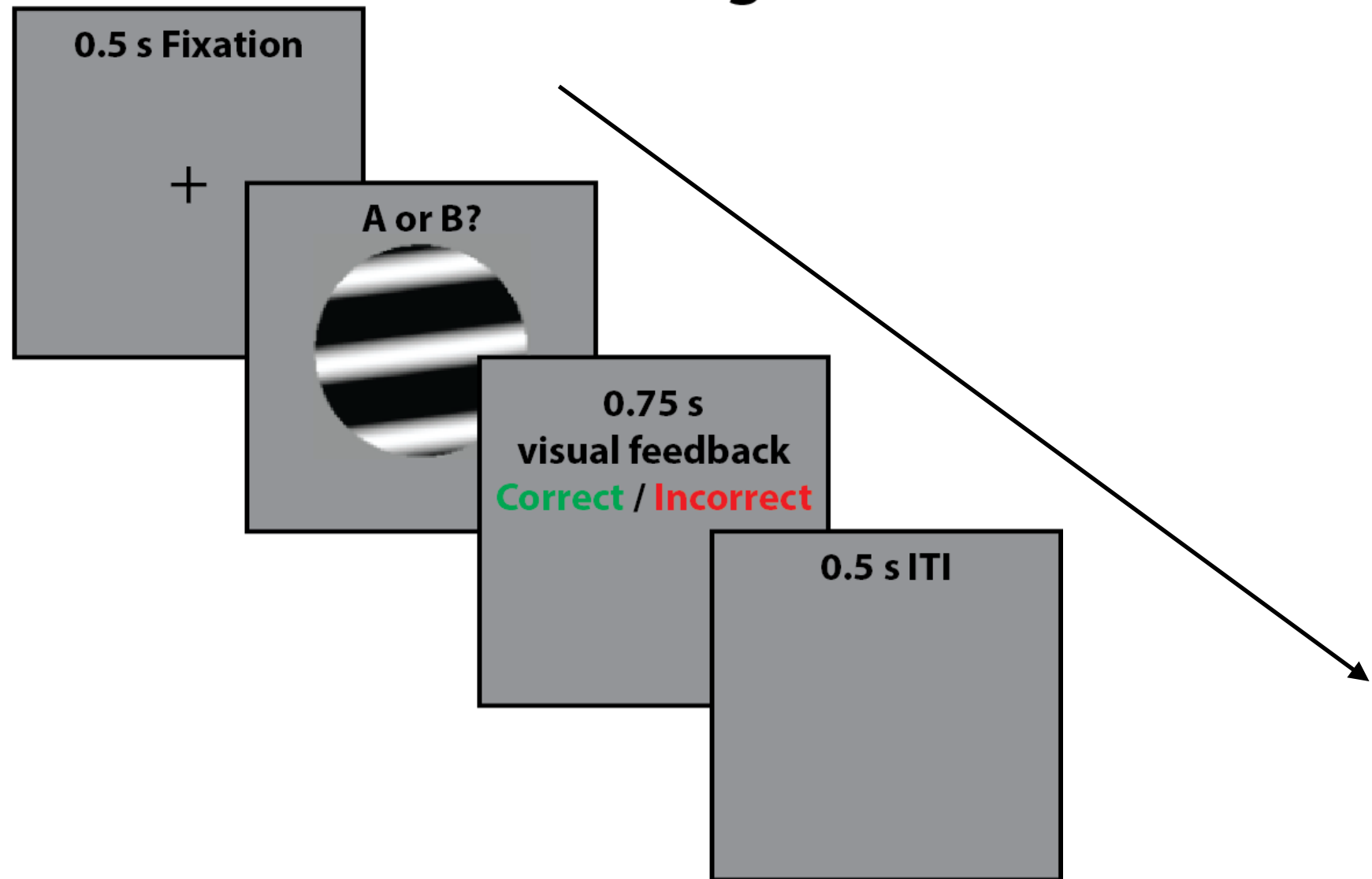
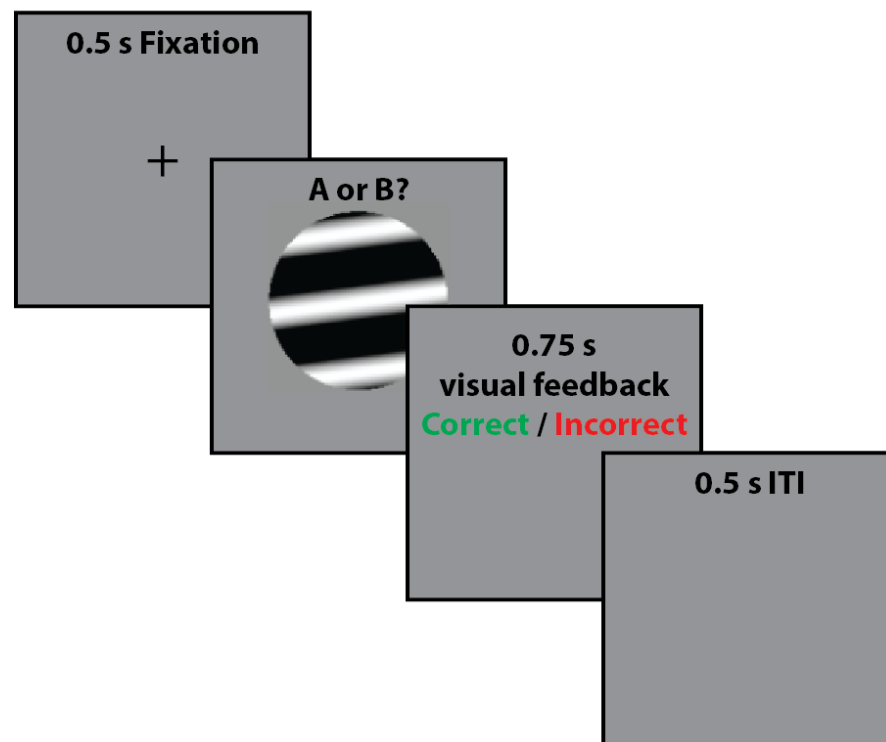


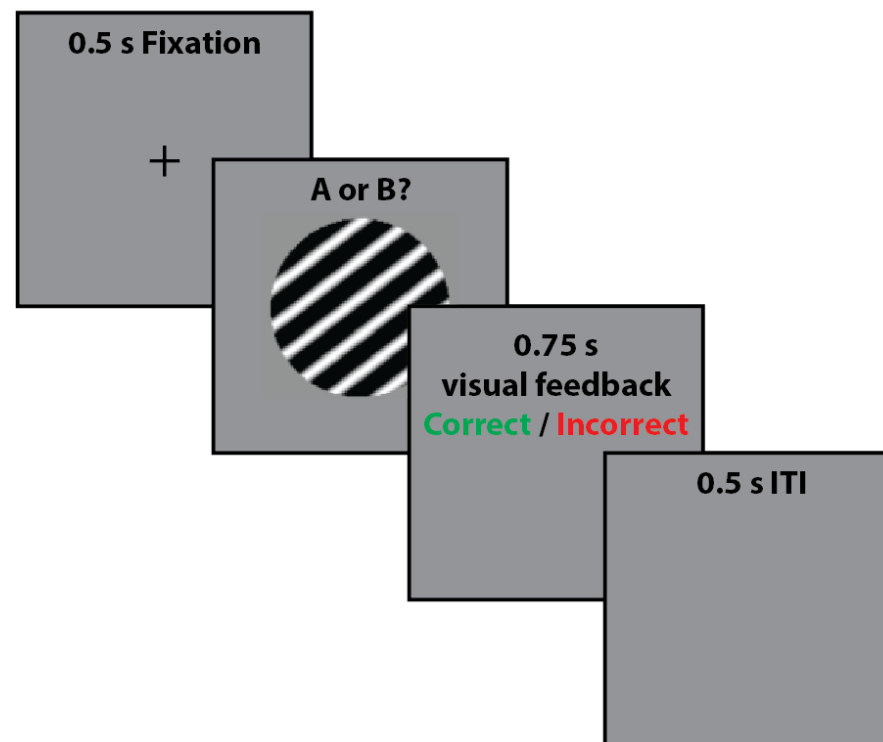
Basic Trial Design



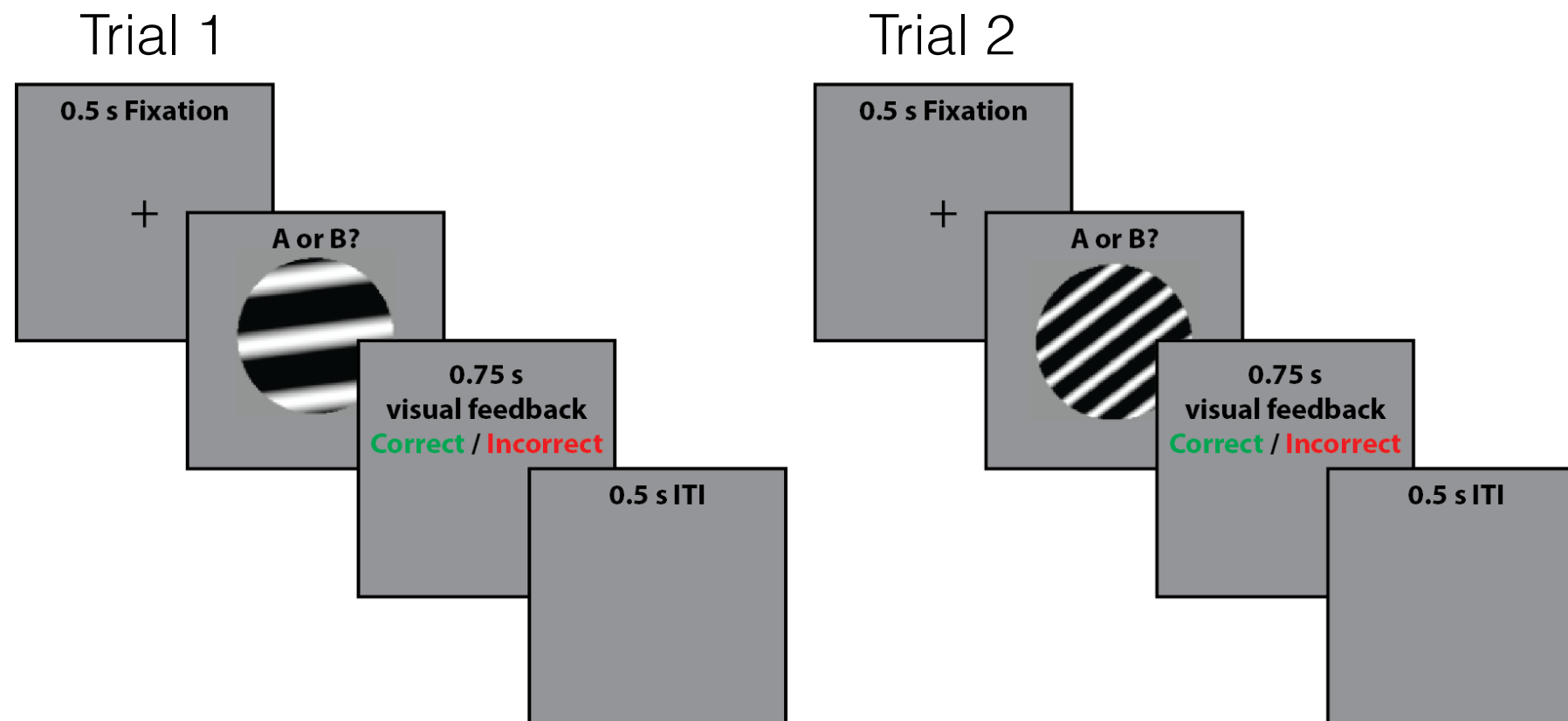
Trial 1



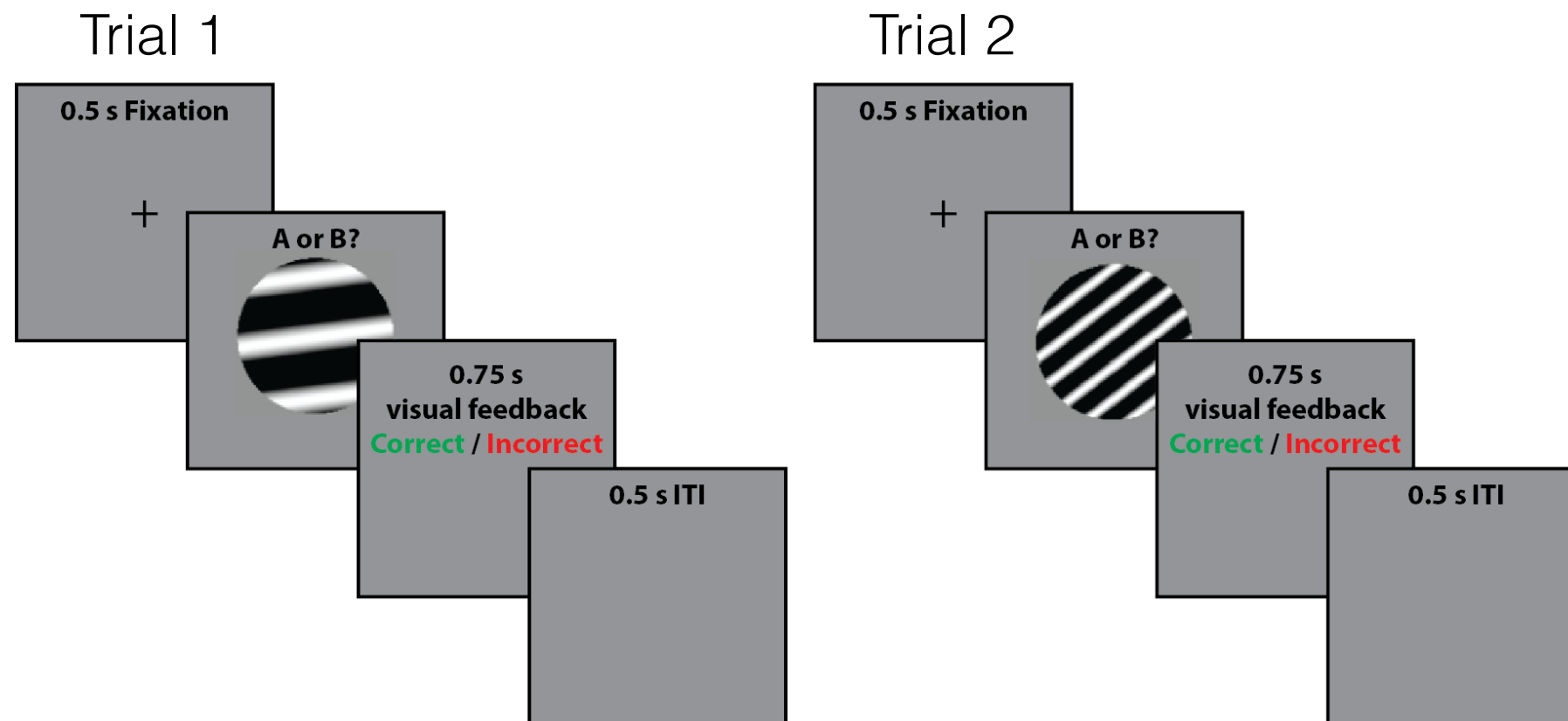
Trial 2



How is this anything like a golf swing?

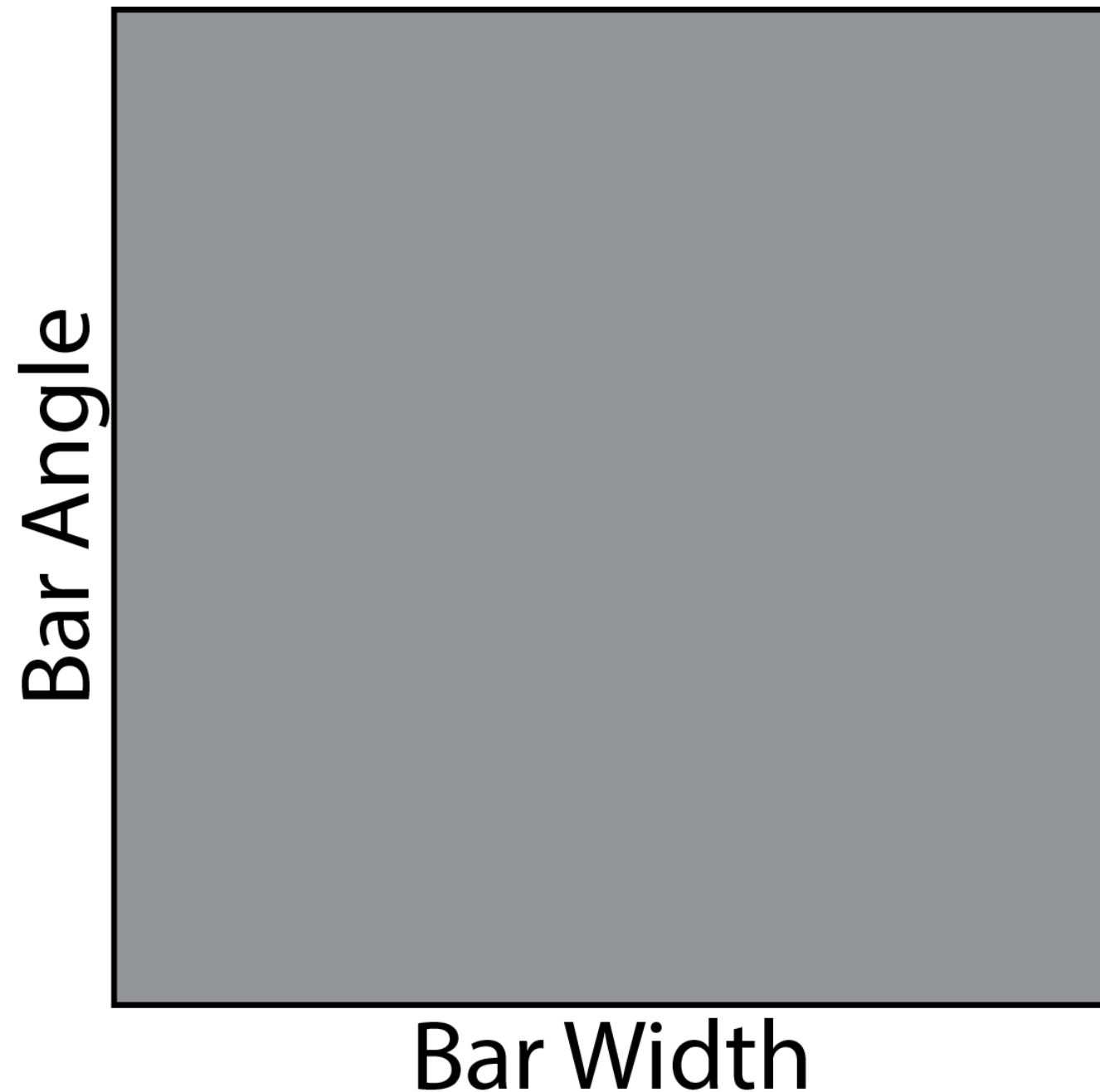


How is this anything like a golf swing?

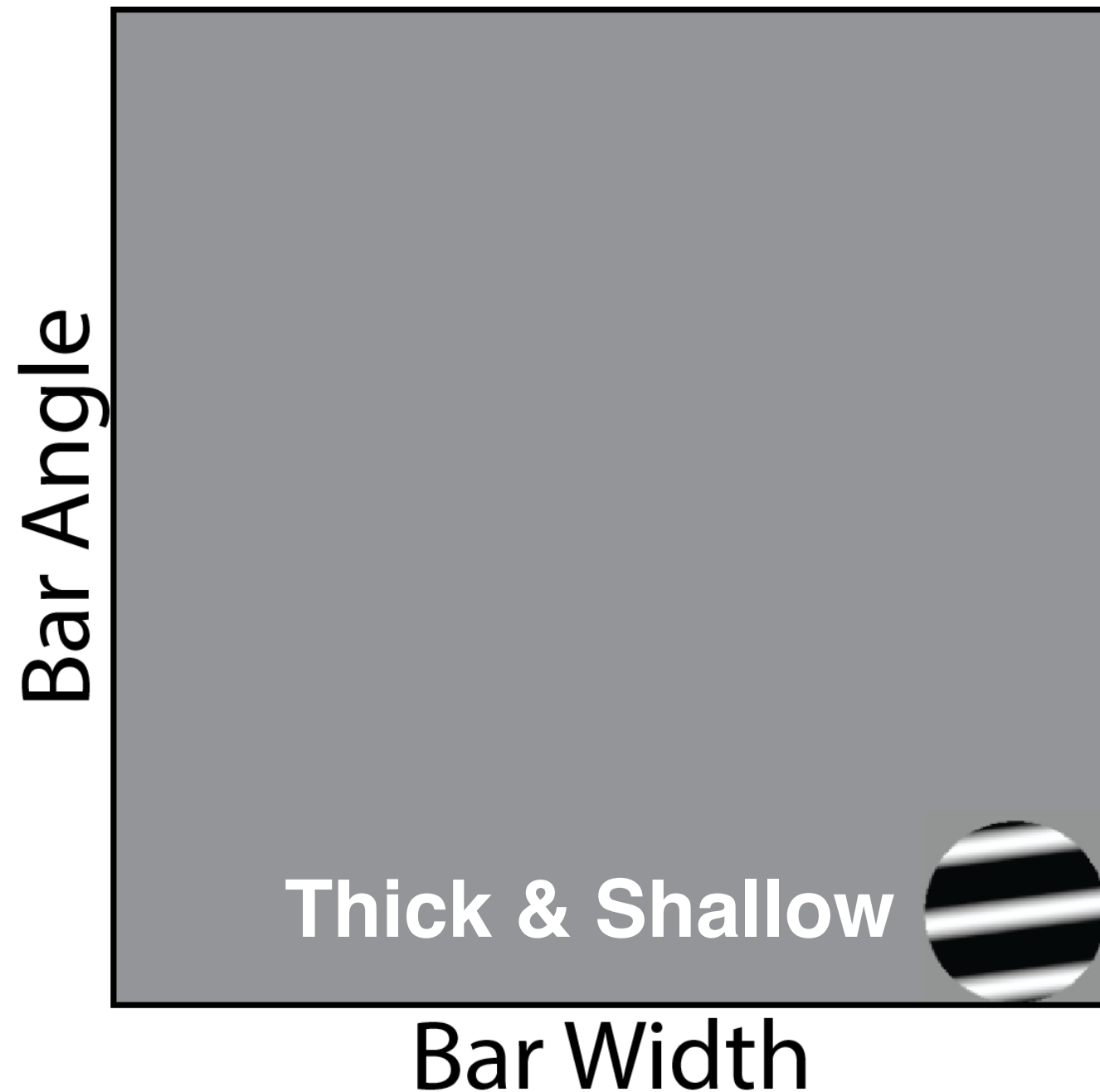


Some variants of this task load skill learning brain networks

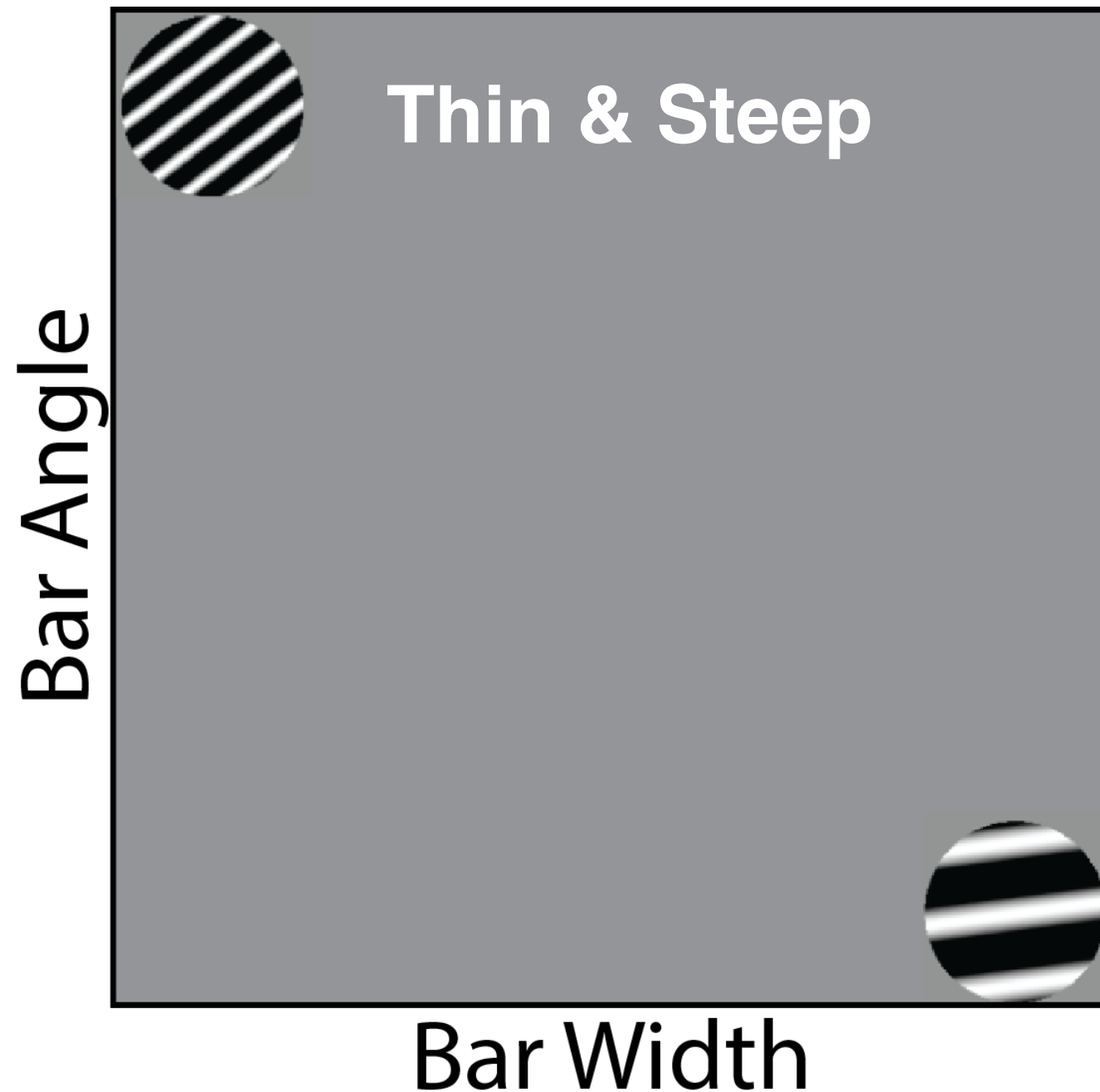
Where do stimuli come from?



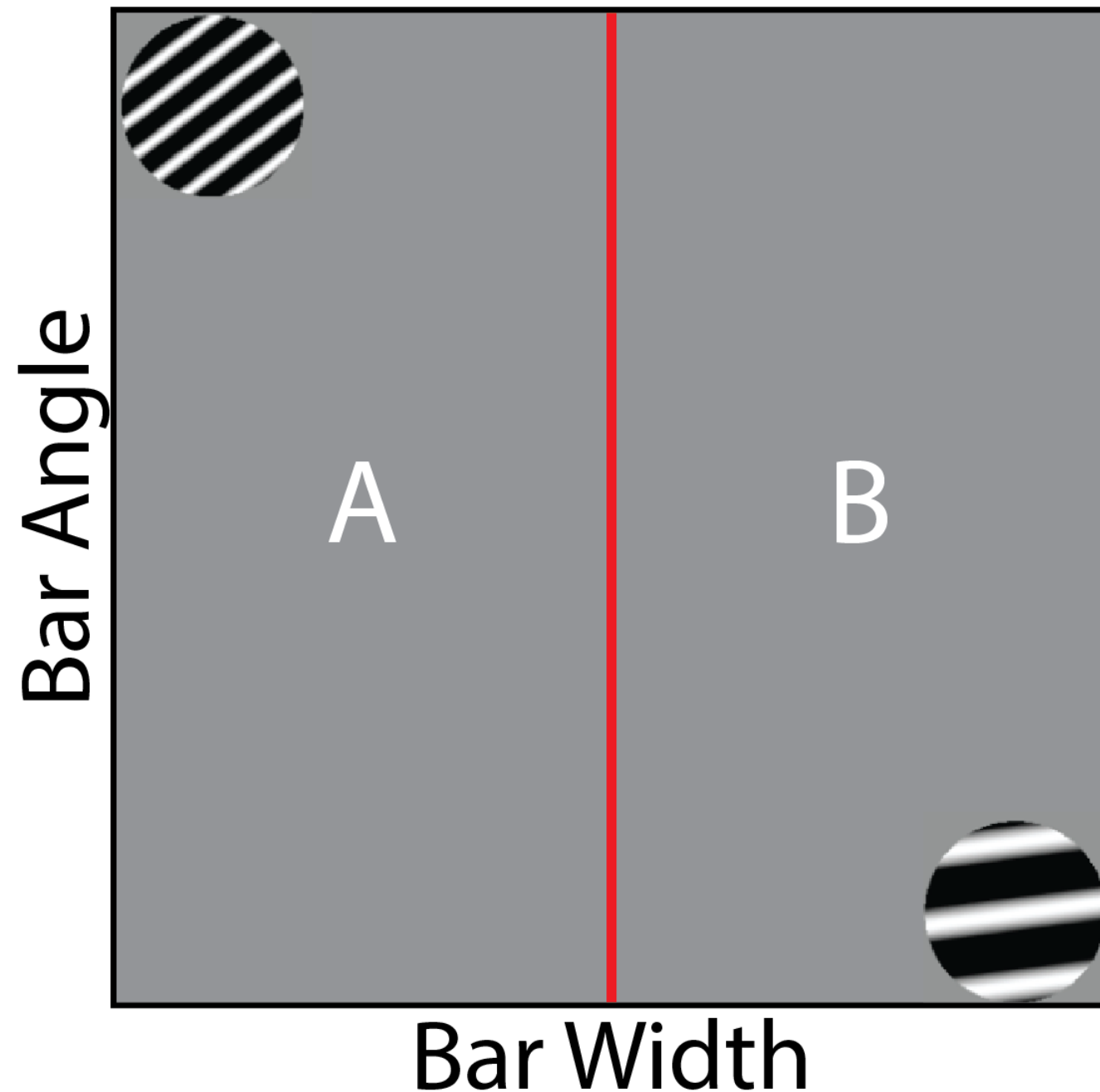
Where do stimuli come from?



Where do stimuli come from?

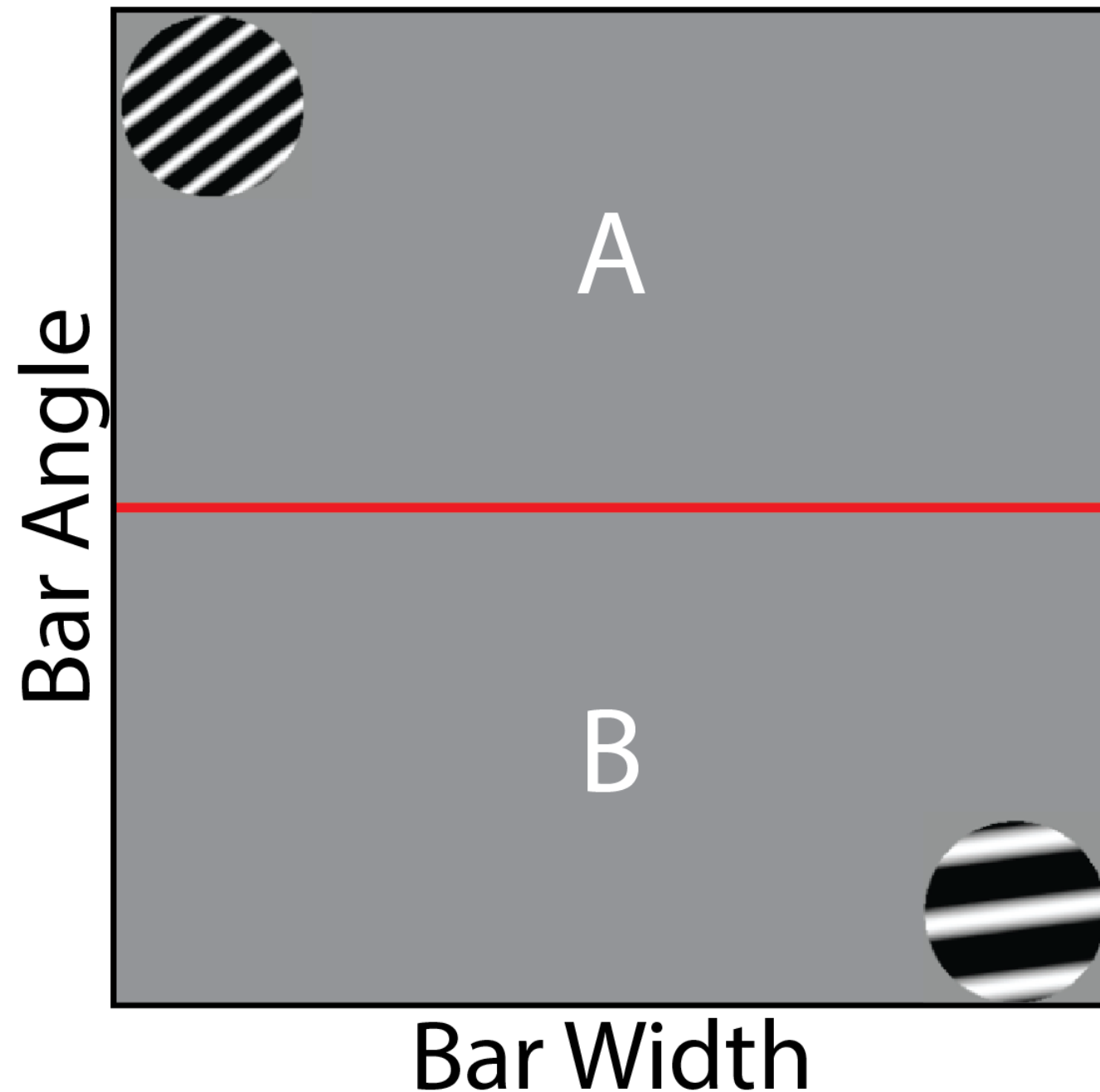


Rule-Based (RB) Category Structures



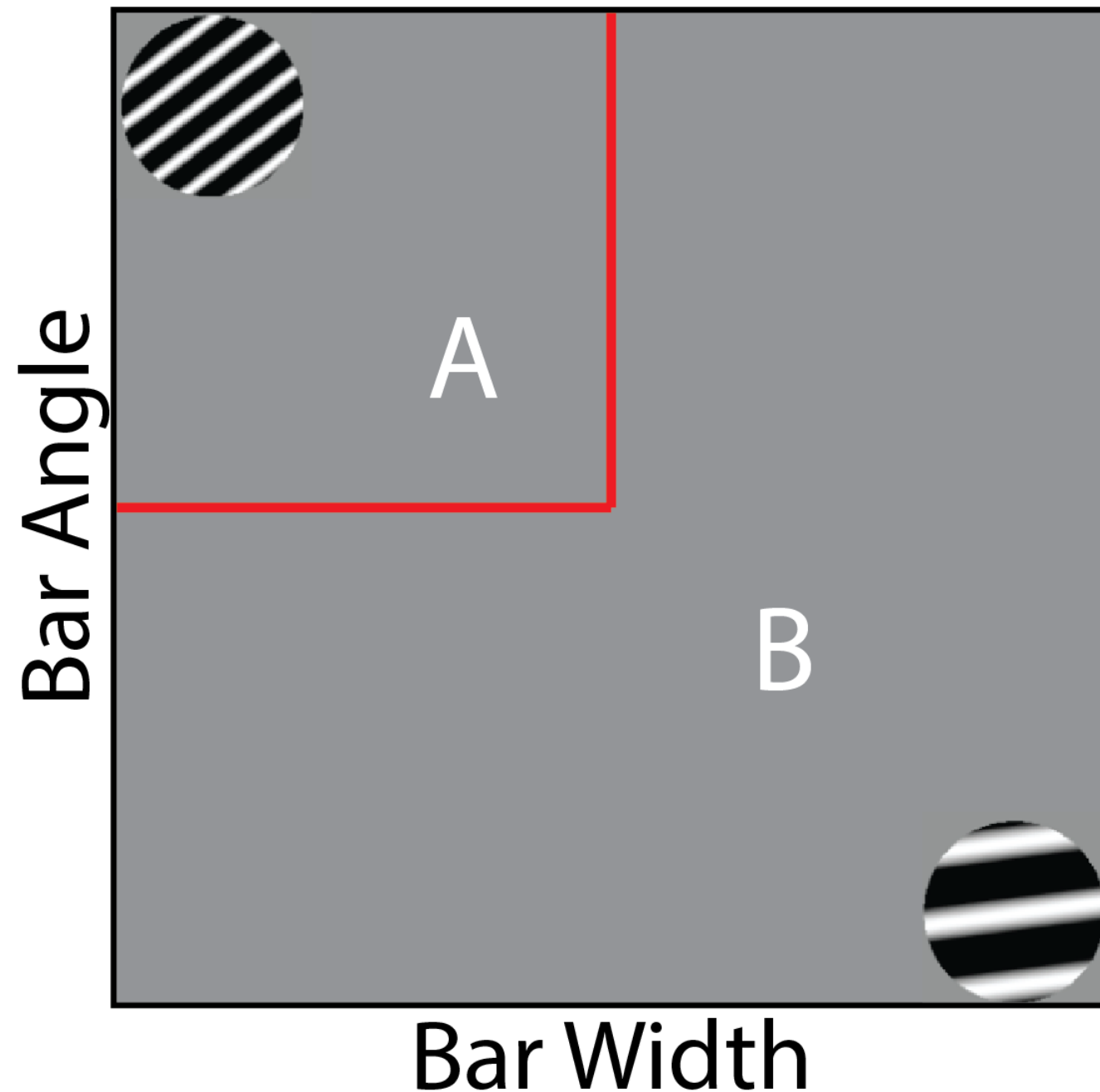
Respond A if bars are thin
Respond B if bars are thick

Rule-Based (RB) Category Structures



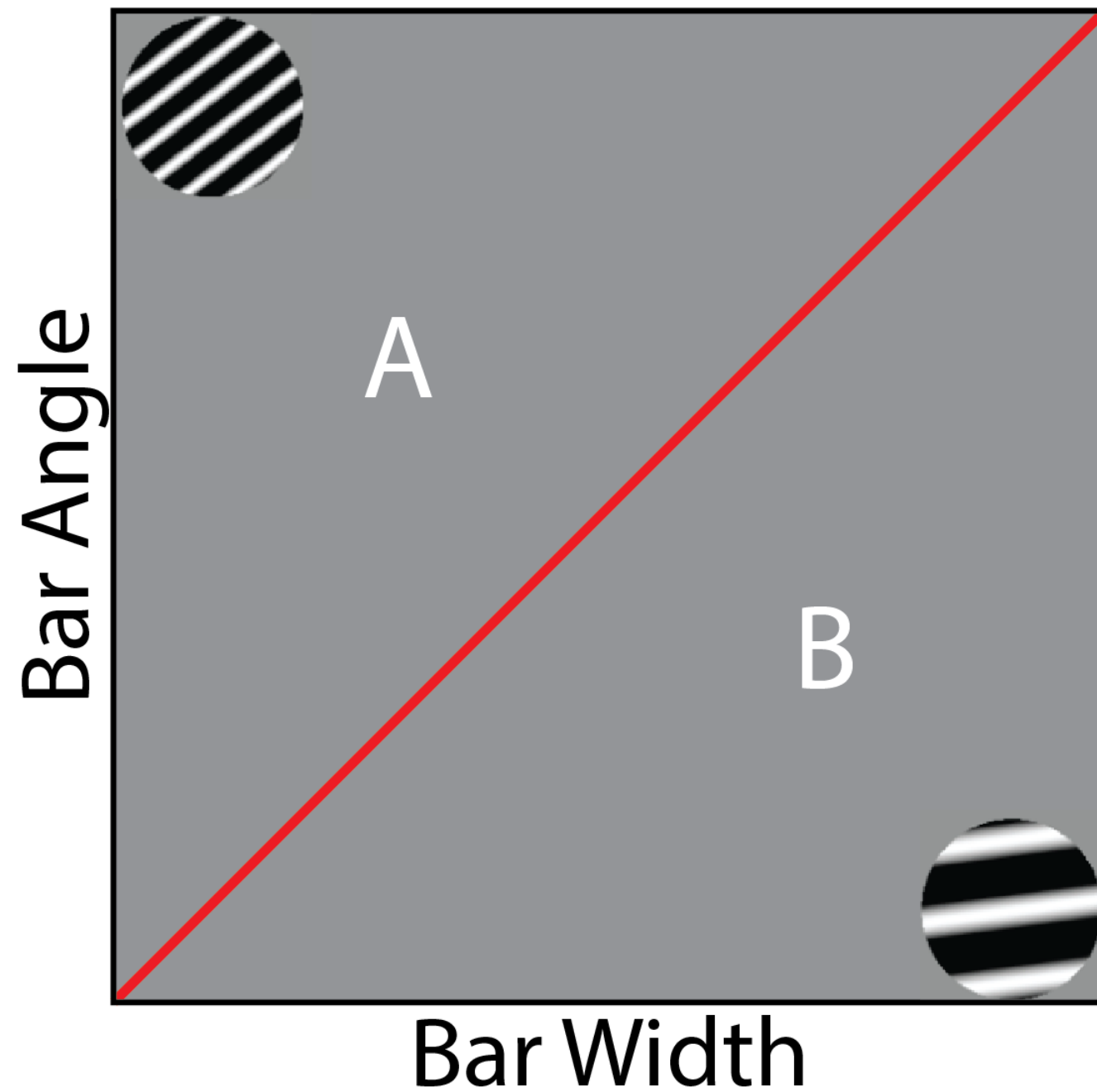
Respond A if bars are steep
Respond B if bars are shallow

Rule-Based (RB) Category Structures

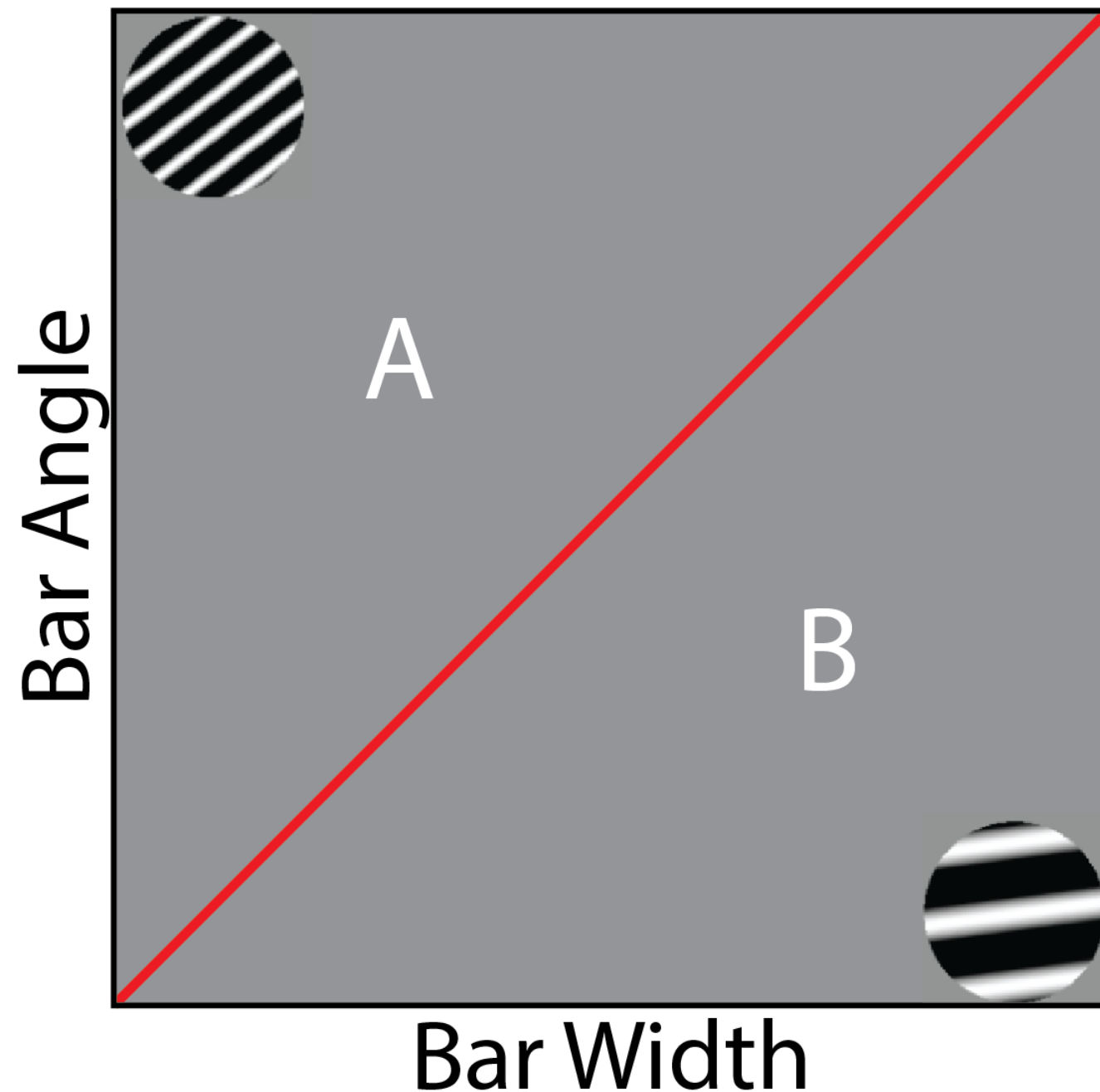


Respond A if bars are thin AND steep
Respond B otherwise

Information-Integration (II) Category Structures

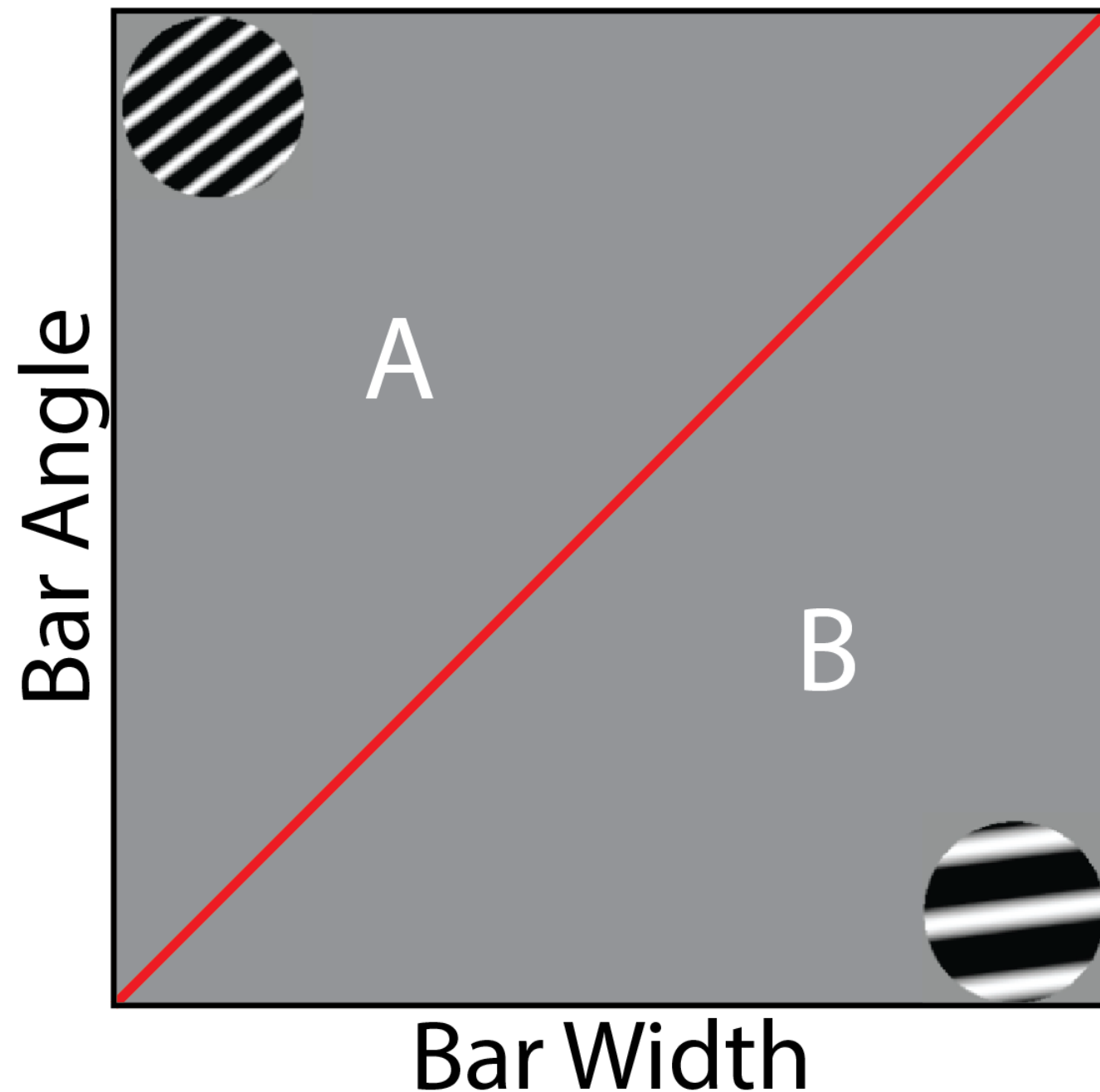


Information-Integration (II) Category Structures



Respond A if bars are steeper than they are thick
Respond B otherwise

Information-Integration (II) Category Structures

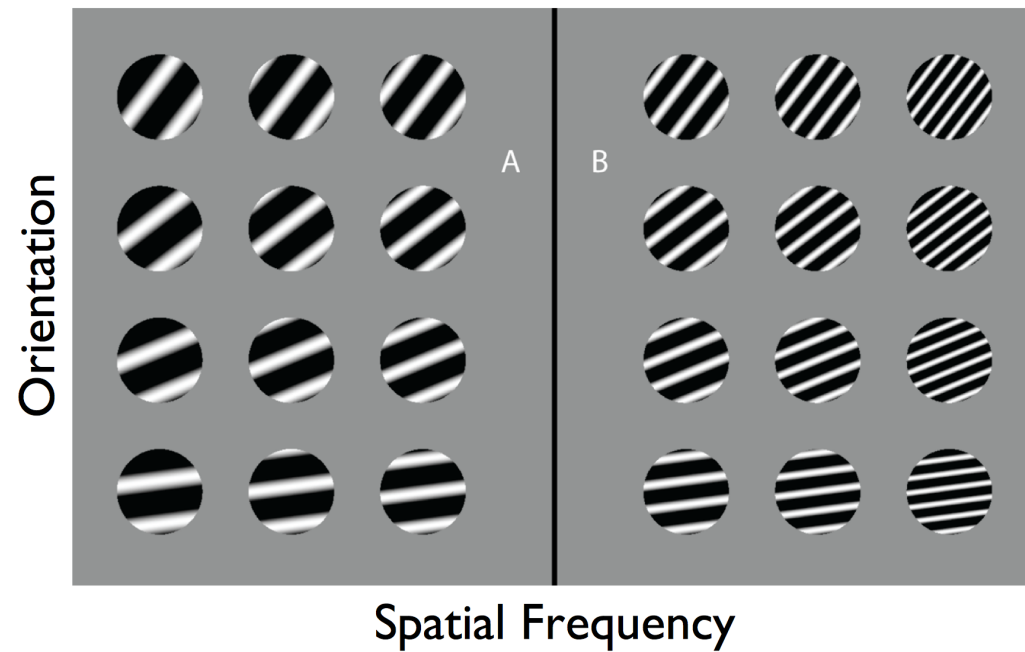


Respond A if bars are steeper than they are thick

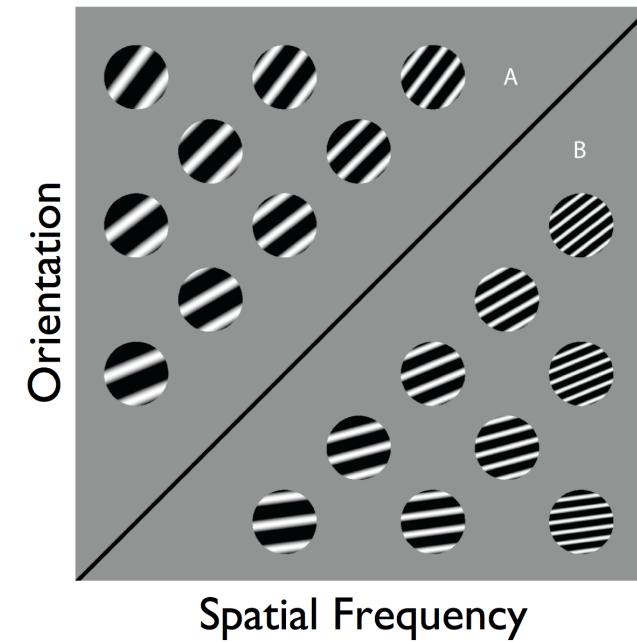
Respond B otherwise

Many dissociations between RB and II category structures

Rule-Based

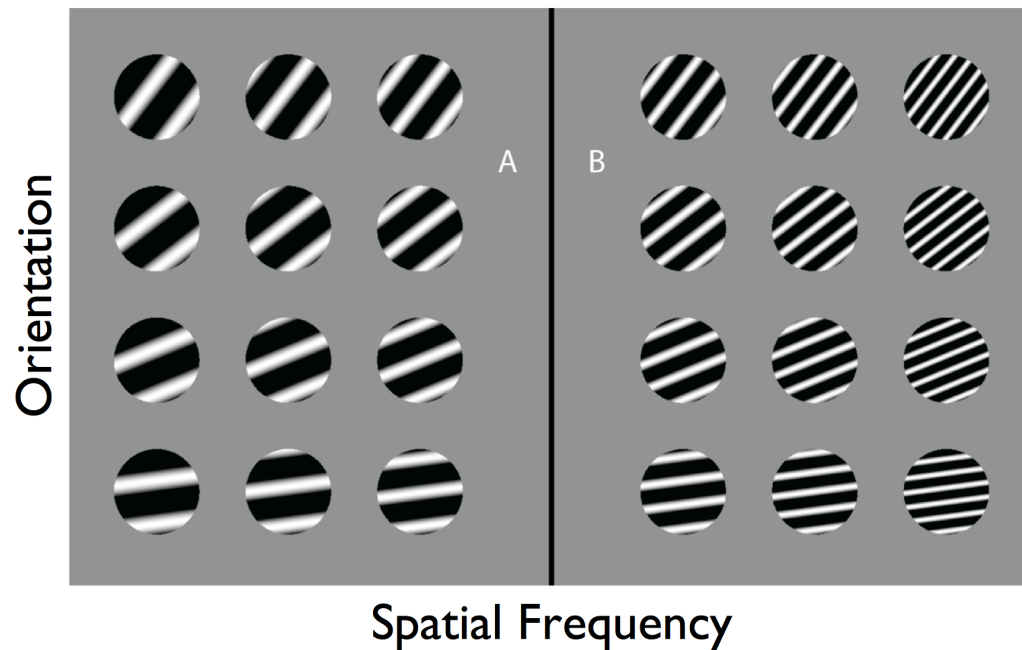


Information-Integration



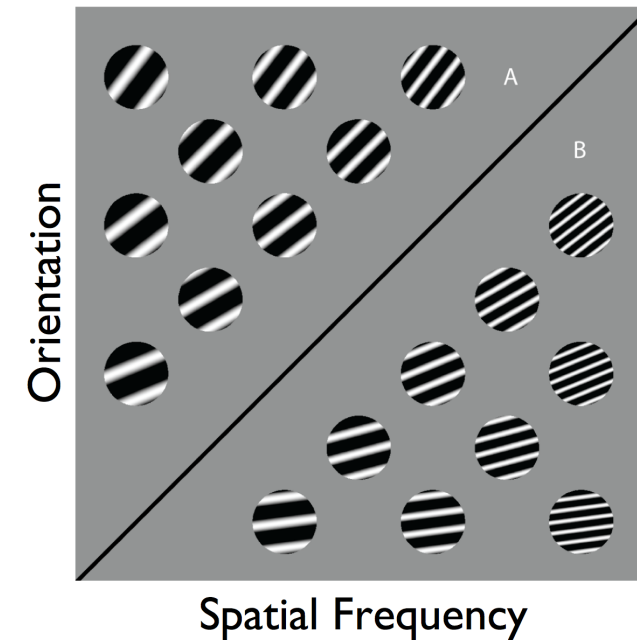
Many dissociations between RB and II category structures

Rule-Based



- Fast acquisition and flexible response remapping

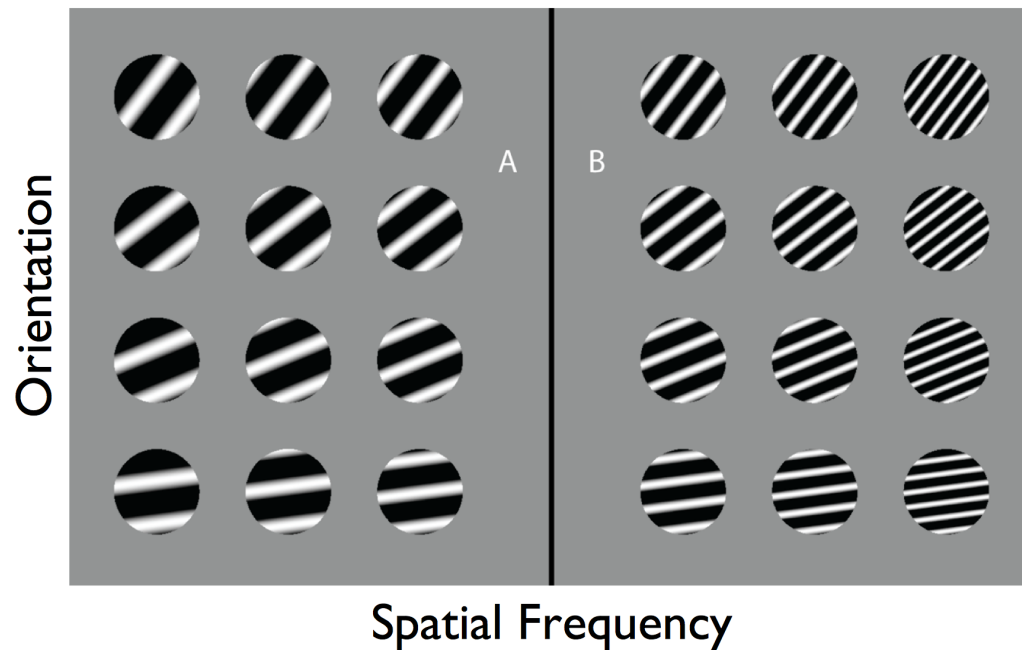
Information-Integration



- Slow acquisition and inflexible response remapping

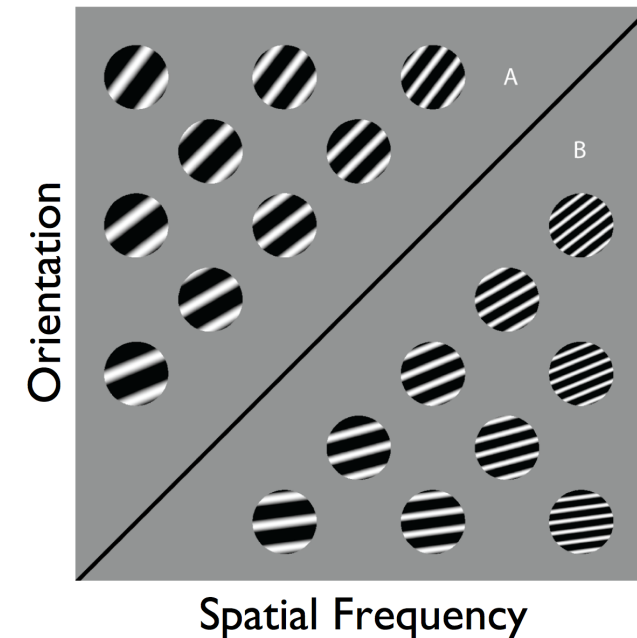
Many dissociations between RB and II category structures

Rule-Based



- Fast acquisition and flexible response remapping
- Massive interference with increased cognitive load

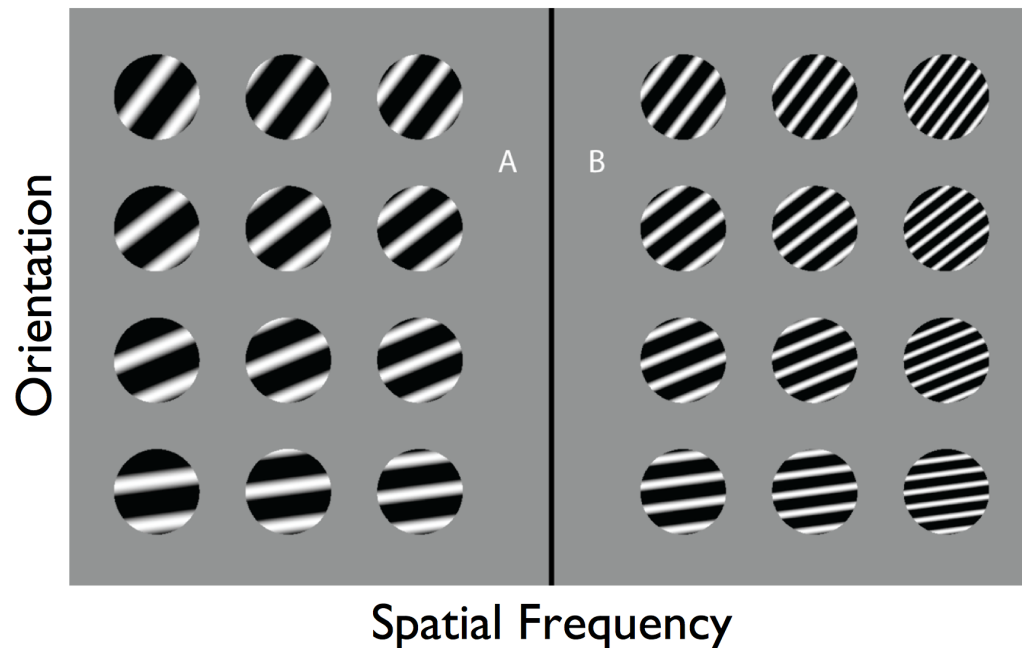
Information-Integration



- Slow acquisition and inflexible response remapping
- Weak interference with increased cognitive load

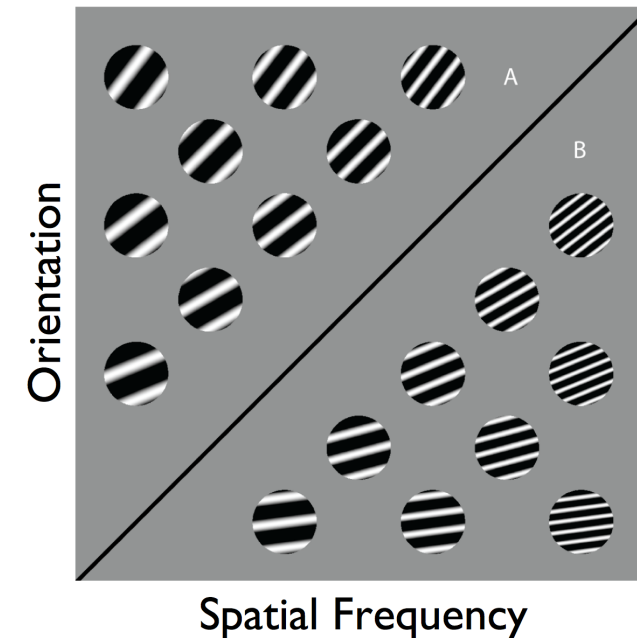
Many dissociations between RB and II category structures

Rule-Based



- Fast acquisition and flexible response remapping
- Massive interference with increased cognitive load
- Learned through explicit hypothesis testing

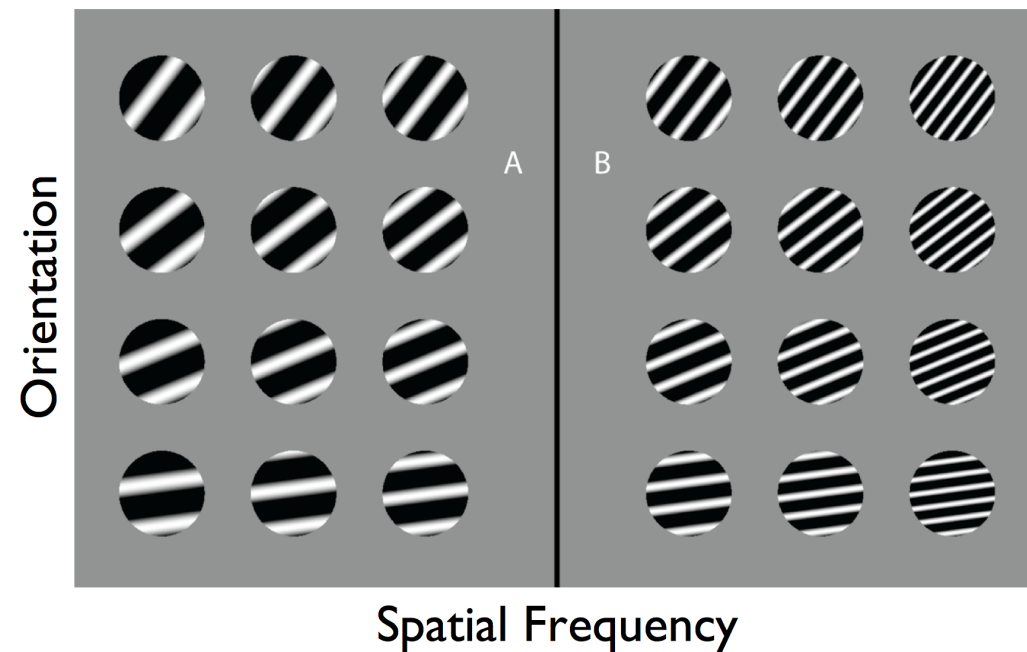
Information-Integration



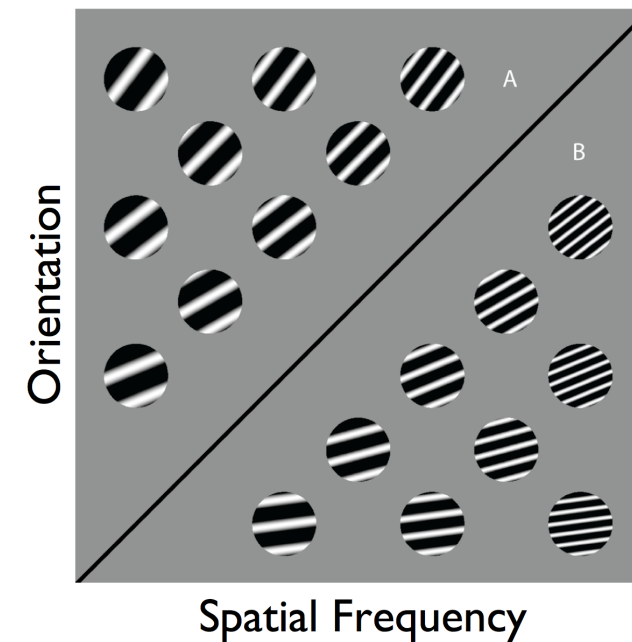
- Slow acquisition and inflexible response remapping
- Weak interference with increased cognitive load
- Learned gradually through trial and error

Different category structures differentially load different memory systems

Rule-Based



Information-Integration



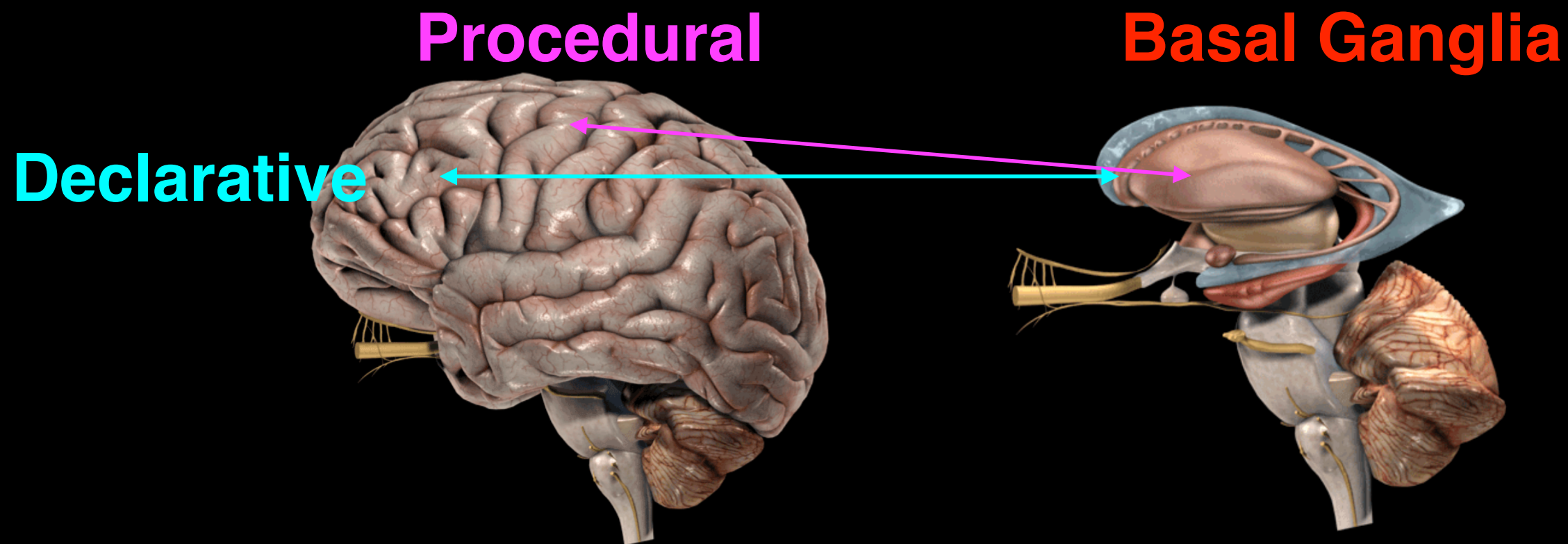
Declarative



Procedural



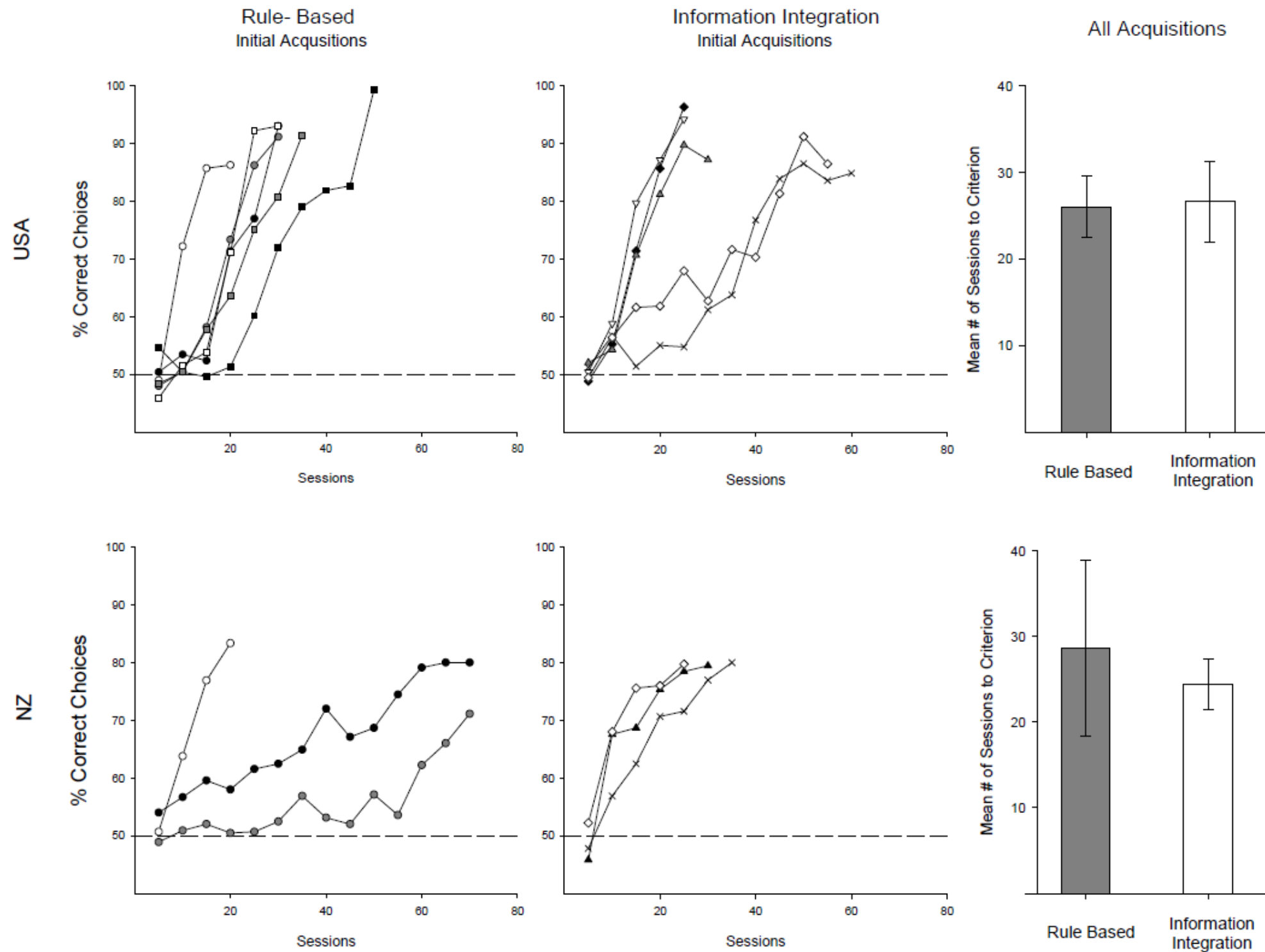
Different memory systems served by distinct neural networks



- Single-cell recording studies
- Animal lesion experiments

- Neuropsychological patient studies
- Neuroimaging experiments

Pigeons (no PFC) learn RB and II equally well

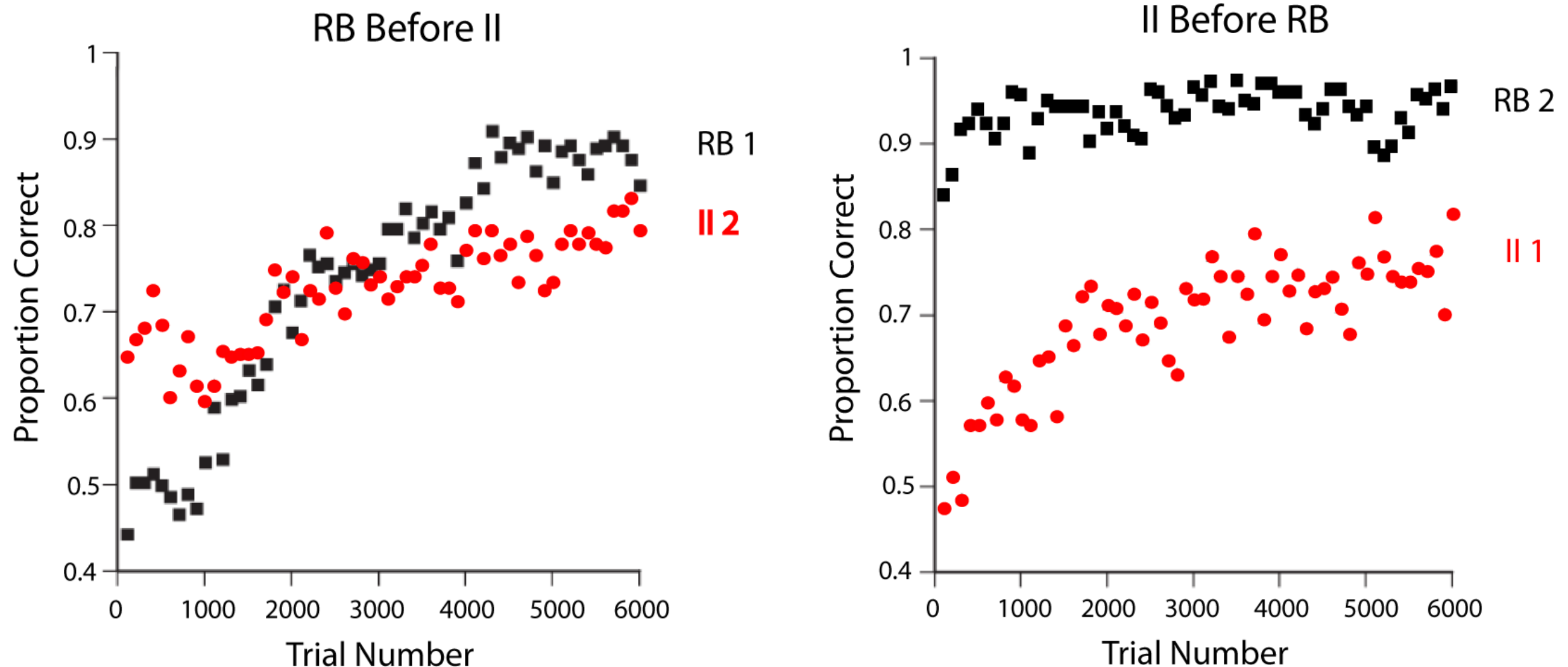


Smith, Ashby, Berg, Murphy, Spiering, Cook, & Grace (2011). PB&R.

Smith, J.D., Berg, M.E., Cook, R.G., Murphy, M. S., Crossley, M.J., Boomer, J., et al. (2012). Neuroscience & Biobehavioral Reviews

Macaques dissociate RB and II like humans

3 animals in each condition

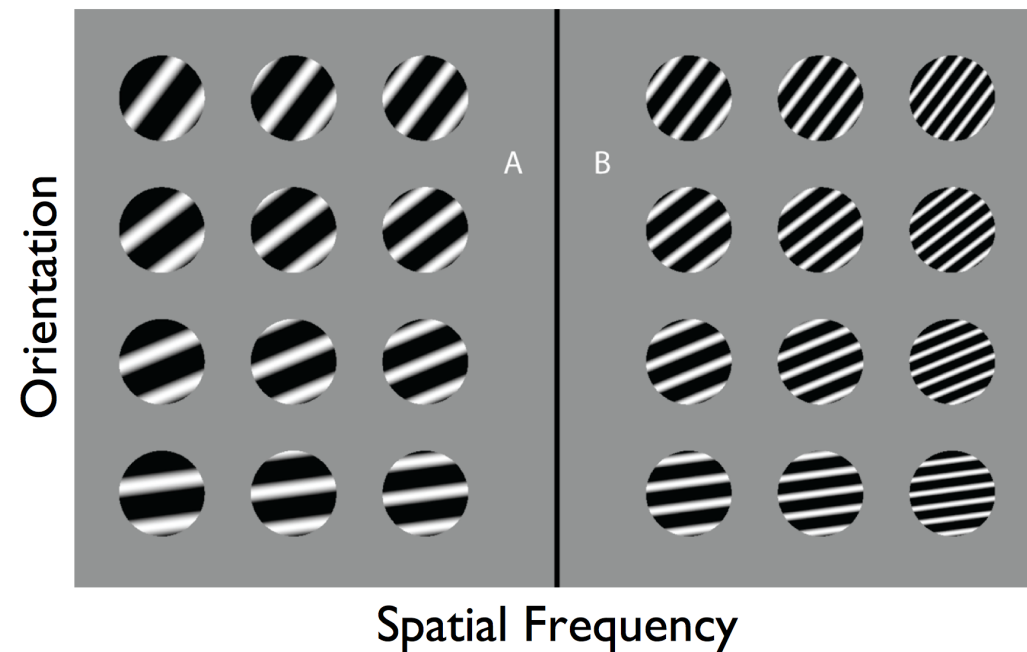


Smith, Beran, Crossley, Boomer, & Ashby (2009). *JEP:ABP*.

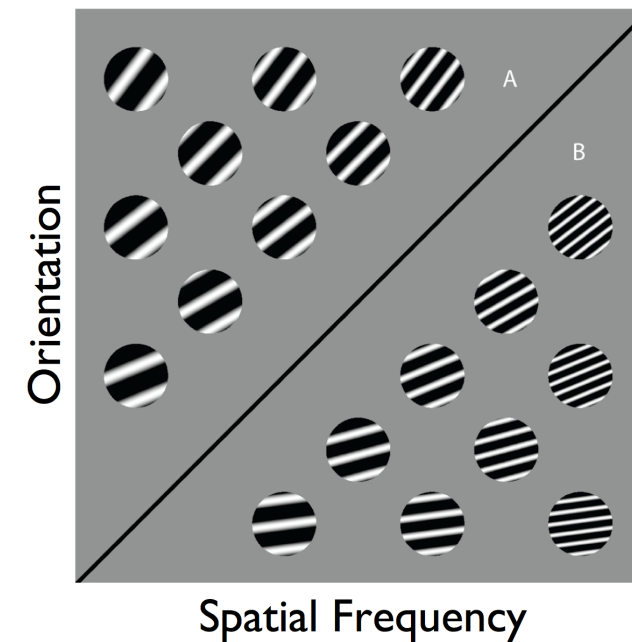
Smith, J.D., Berg, M.E., Cook, R.G., Murphy, M. S., Crossley, M.J., Boomer, J., et al. (2012). *Neuroscience & Biobehavioral Reviews*

Different category structures differentially load different memory systems

Rule-Based



Information-Integration



Declarative



Procedural

